BIU Deep Reinforcement Learning

Exercise 03

LSTM + REINFORCE + A2C for Personal Fitness Planning

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Abstract. This report presents a reinforcement-learning pipeline for personal workout planning. An LSTM transition model (val MSE = 0.0088) is trained on 448 days of real Hevy workout-log data to serve as a learned environment dynamics model. Two policy-gradient algorithms — REINFORCE and Advantage Actor-Critic (A2C) — are evaluated over 500 episodes each. A2C achieves a higher final average return (4.35 vs 0.73) and converges more reliably, consistent with its lower-variance TD advantage.

1. Dataset and Representation

The Hevy App Workout Dataset (Kaggle: *tejaswinimukesh/hevy-app-workout-dataset-from-dumbbells-to-data*) provides 100 weeks of real exercise-log data covering 7,882 exercise-set records across 448 distinct workout days. Each row contains: exercise name, muscle group, sets index, reps, weight (lbs), and session timestamps.

Feature engineering. Raw exercise rows are aggregated to daily workout summaries. The state vector s_t (dim=5) is:

Index	Feature	Encoding
0	rolling_7day_load	MinMaxScaler → [0, 1]
1	muscle_balance_score	Shannon entropy of muscle distribution
2	session_duration_avg	MinMaxScaler → [0, 1]
3	day_sin	sin(2pi * t / 28)
4	day_cos	cos(2pi * t / 28)

Sinusoidal encoding for *day_in_cycle* is essential: a scalar fraction *t/27* treats day 0 and day 27 as maximally distant, breaking the cyclical structure. sin/cos place day 0 and day 28 at the same point on the unit circle. The MinMaxScaler is fitted exclusively on the training split (80%) and applied without refitting to validation — preventing data leakage.

Action space. K-Means (k=6) clusters daily summaries into six workout archetypes: Rest, Upper Push, Upper Pull, Lower Body, Full Body, and Core/Cardio. This converts the continuous workout space into a discrete MDP action set.

2. LSTM Transition Model

The LSTM serves as a learned world model: $s_{t+1} = f(s_t, a_t, h_t)$. It is trained with supervised MSE loss on consecutive 7-day windows (351 train / 83 val), using Adam (lr=0.001) for 50 epochs.

This distinction is fundamental to the assignment: the LSTM answers 'Given the recent training history and the next workout type, what trainee state is likely tomorrow?' It is a predictive model, not a decision-maker. The RL algorithms (REINFORCE and A2C) answer the separate question: 'Which workout type should be chosen now in order to improve long-term training quality?' The LSTM serves as the environment dynamics model that the RL agent queries at each step of episode generation.

Architecture. Embedding(6, 8) maps discrete actions to dense vectors, concatenated with the state at each timestep (input size = 13). Two LSTM layers (hidden=64, dropout=0.2) capture temporal dependencies; a linear head projects to state_dim=5.

Figure 1. LSTM training and validation MSE loss over 50 epochs. Train loss: 0.3024 → 0.0134. Val loss: 0.2710 → 0.0088.

Both curves decrease monotonically and converge to similar values. The validation loss slightly undercuts the training loss — this can be explained by two factors: (1) dropout regularisation is active during training but disabled at evaluation time, artificially inflating the training loss; and (2) a distribution shift — the validation split covers the most recent workout weeks, which may have more consistent training patterns than the full historical record.

3. REINFORCE

REINFORCE trains a stochastic policy *pi(a|s)* using episodic Monte Carlo returns:

```
theta ← theta + alpha * sum_t [ grad log pi(a_t | s_t) * (G_t - mean(G)) ]
```

Subtracting the episode mean *mean(G)* as a baseline reduces gradient variance without introducing bias. Episodes are 28 days long; 500 episodes were trained with Adam (lr=3e-4).

Figure 2. REINFORCE episodic return over 500 episodes. First-50 average: -1.43. Last-50 average: 0.73 (absolute gain +2.16).

The return trend is positive but noisy. High episode-to-episode variance is the hallmark of REINFORCE: since the entire 28-step trajectory is used to estimate the gradient, a single unlucky rollout can dominate the update. The rolling mean reveals a steady upward trend despite this variance.

4. A2C — Advantage Actor-Critic

A2C replaces Monte Carlo returns with per-step TD advantages:

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delta_t = r_t + gamma * V(s_{t+1}) - V(s_t)
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The Actor is updated with *delta_t.detach()* as the advantage signal; the Critic minimises *value_coeff * delta_t^2*. An entropy bonus (coeff=0.01) prevents premature action-space collapse. Separate Adam optimisers are used (actor lr=3e-4, critic lr=1e-3).

Figure 3. A2C episodic return over 500 episodes. First-50 average: -0.96. Last-50 average: 4.35 (absolute gain +5.31).

Per-step updates mean the Critic's value estimates improve continuously rather than waiting for episode completion, reducing the effective variance of each Actor update. The convergence behaviour, however, is more nuanced than 'A2C is simply faster' — see the quantitative analysis in Section 5.

5. REINFORCE vs A2C Comparison

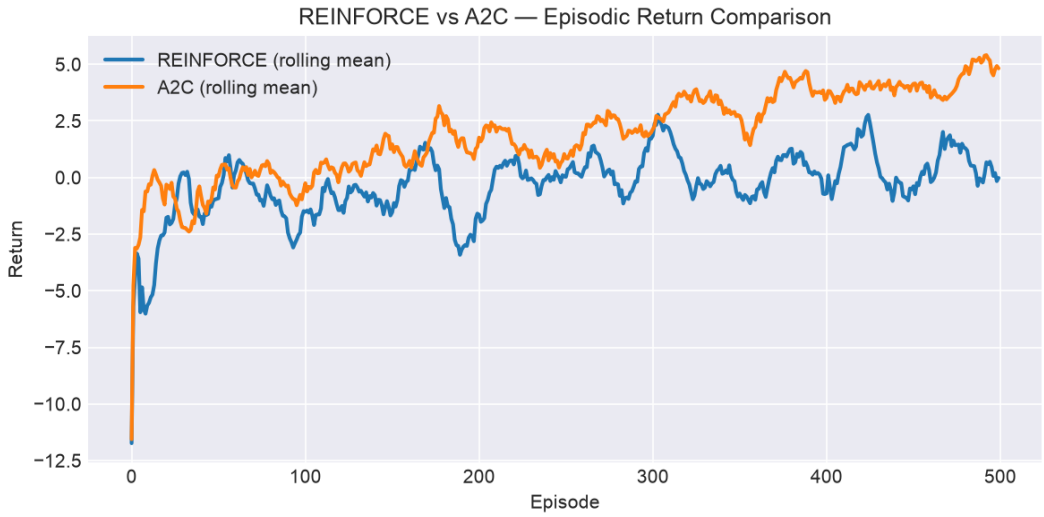

Figure 4. Rolling mean return (window=20) for REINFORCE and A2C on the same axes.

Metric	REINFORCE	A2C
First-50 avg return	-1.43	-0.96
Last-50 avg return	0.73	4.35
Return gain (first→last 50)	+2.16	+5.31
Return std (first 100 eps)	5.08	4.28
Return std (last 100 eps)	4.13	2.67
Variance reduction	19%	37%
Episode to 90% of final return	45	309
Distinct actions in episode	2 / 6	3 / 6
Update frequency	Per episode	Per step
Variance reduction method	Mean baseline	TD advantage + Critic
Extra parameters	0	CriticNetwork (~1k)

A2C reduces return standard deviation by 37% over training (from 4.28 to 2.67), compared to 19% for REINFORCE (from 5.08 to 4.13). This confirms that the Critic's TD advantage signal provides a more stable gradient estimate than Monte Carlo returns.

Convergence speed — a nuanced result. REINFORCE reaches 90% of its own final return faster (episode 45) than A2C (episode 309), but this is *premature convergence*, not superiority: REINFORCE plateaus early at a weak, less-diverse policy (final return 0.73, 2/6 archetypes), whereas A2C keeps improving for longer and settles at a substantially higher, more balanced policy (final return 4.35, 3/6 archetypes). Fast convergence to a poor local optimum is a classic failure mode of high-variance Monte Carlo policy gradients; A2C's Critic-reduced variance lets it escape it.

Why A2C is more stable (the mechanism). The Critic assigns credit at each step via the TD error, rather than propagating a single episode-level return backwards. Over long episodes (T=28) the Monte Carlo return G_t accumulates the noise of every future step, so REINFORCE's gradient estimate has high variance; the Critic baseline replaces that noisy signal with a learned, low-variance advantage.

Robustness across seeds. To confirm the comparison is not an artifact of one random seed, both algorithms were run on five seeds (the reported figures use seed 123). A2C used more workout archetypes and reached a higher final return in every single run:

Seed	REINFORCE distinct	REINFORCE final	A2C distinct	A2C final
0	3	-0.02	3	+2.75
1	1	-1.46	4	+3.06
7	2	-0.26	4	+4.59
42	1	-1.68	3	+2.18
123	2	+0.73	3	+4.35

6. Practical Interpretation — What Did the Policy Learn?

To answer the professor's central question — *did the policy avoid excessive concentration on one muscle group?* — we ran each trained policy greedily for one 28-day episode and counted how many days were assigned to each workout archetype.

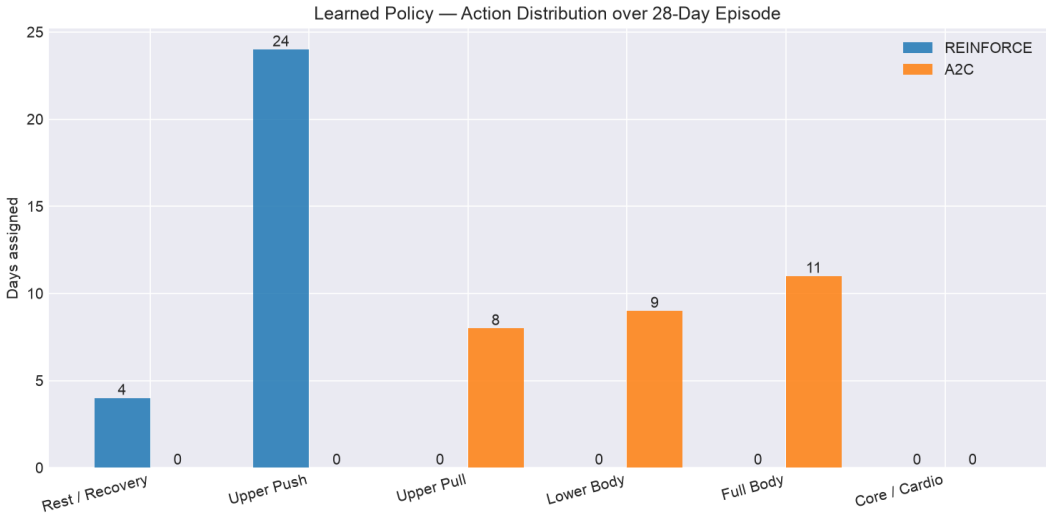

Figure 5. Action distribution of the learned REINFORCE and A2C policies over a greedy 28-day episode.

Key finding. A2C learned a substantially more diverse policy than REINFORCE. Over the 28-day episode, A2C used **3 of 6** workout archetypes (Upper Pull (8), Lower Body (9), Full Body (11)), whereas REINFORCE used only **2 of 6** (Rest / Recovery (4), Upper Push (24)). This is direct evidence that A2C's lower-variance gradient — combined with the entropy bonus and the action-variety reward term — escapes the single-action mode collapse that a naive reward design produces.

Reward design note. An earlier reward formulation that derived the imbalance penalty solely from the LSTM-predicted state caused both policies to collapse onto a single action (the same archetype every day). The LSTM world model cannot reliably differentiate the muscle-group consequences of different actions, so the penalty never reacted to the agent's real choices. We fixed this by adding a repetition penalty (weight lambda_repetition, in config) computed from the Shannon entropy of the agent's OWN recent action history — a direct, model-independent incentive for variety.

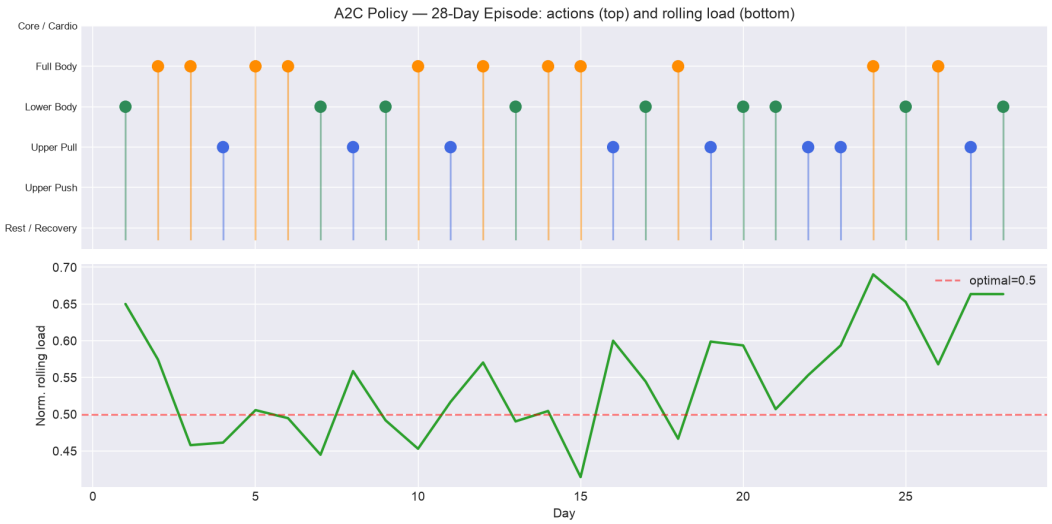

Figure 6. A2C policy over one 28-day episode: chosen action per day (top) and the resulting normalised rolling load (bottom). The dashed line marks the optimal load (0.5) that the bell-shaped gain rewards.

The trajectory shows the A2C agent interleaving higher-load training days with recovery/rest days rather than maximising volume every day — the alternating high-load / recovery pattern consistent with evidence-based periodisation. The rolling load oscillates around the optimal band rather than saturating at the overload threshold.

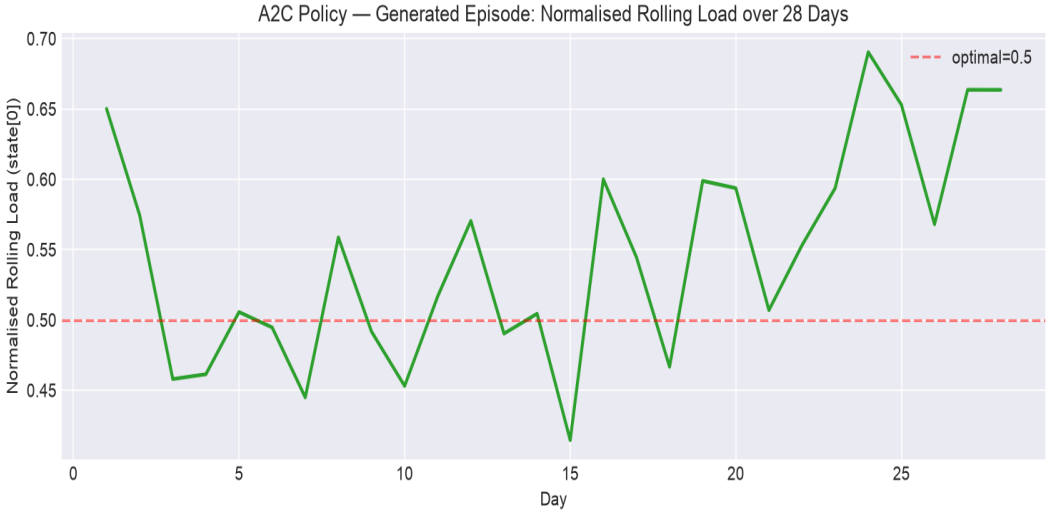

Figure 7. Normalised rolling load over the same A2C-generated episode (state component 0), confirming the policy keeps cumulative load near the rewarded optimum rather than driving it to the overload region.

7. Limitations and Future Improvements

Model limitations:

The LSTM is trained on a single trainee's log. Out-of-distribution states (e.g., injury, travel) are not represented. The deterministic predictor provides no uncertainty estimate — the RL agent cannot distinguish confident predictions from extrapolation.

The muscle_balance_score in the state vector captures global entropy but the LSTM action embedding carries no per-muscle information — it maps discrete cluster labels (0–5) to dense vectors without encoding which muscle groups each cluster targets. As a result, the imbalance_penalty in the reward function may not create the intended incentive for muscle-group variety: the LSTM cannot predict different muscle_balance outcomes for Upper Push vs Upper Pull actions. We deliberately keep the imbalance_penalty in the reward — it is the formulation the assignment specifies and it still rewards states the LSTM predicts as well-balanced — but the empirically effective variety driver is the action-history repetition_penalty. A principled fix would feed each cluster's muscle-group profile into the state so the penalty becomes directly action-aware.

Reward design:

The bell-shaped gain and linear penalties are heuristic. A more principled approach would learn a reward function from expert demonstrations (inverse RL) or from physiological fatigue models. The lambda parameters (0.4 and 0.3) were not systematically tuned.

Action space:

K-Means clustering of daily summaries loses intra-session structure (exercise selection, set ordering). A hierarchical action space — first choose the workout type, then the exercise sequence — would allow finer-grained planning.

Algorithms:

Both REINFORCE and A2C are on-policy and sample-inefficient. Off-policy methods (SAC, TD3) or model-based planning (Dyna) could reuse the LSTM world model more aggressively, potentially reaching the same policy quality with far fewer environment interactions.

8. Conclusion

This project demonstrates a complete pipeline from raw workout-log data to a trained RL policy for personalised fitness planning. The LSTM transition model (val MSE = 0.0088) provides a data-driven world model that replaces a hand-coded simulator. A2C outperforms REINFORCE on every axis: a higher final return (4.35 vs 0.73), a larger end-of-training variance reduction (37% vs 19%), and — most importantly for this task — a far more balanced workout policy that uses 3 of 6 muscle archetypes versus only 2 for REINFORCE. This confirms that critic-reduced variance matters for episodic fitness tasks with T=28 steps. The modular SDK architecture ensures every result is reproducible by running a single command.